

TOPIC: AI-Powered Digital Forensics & Investigation Assistant

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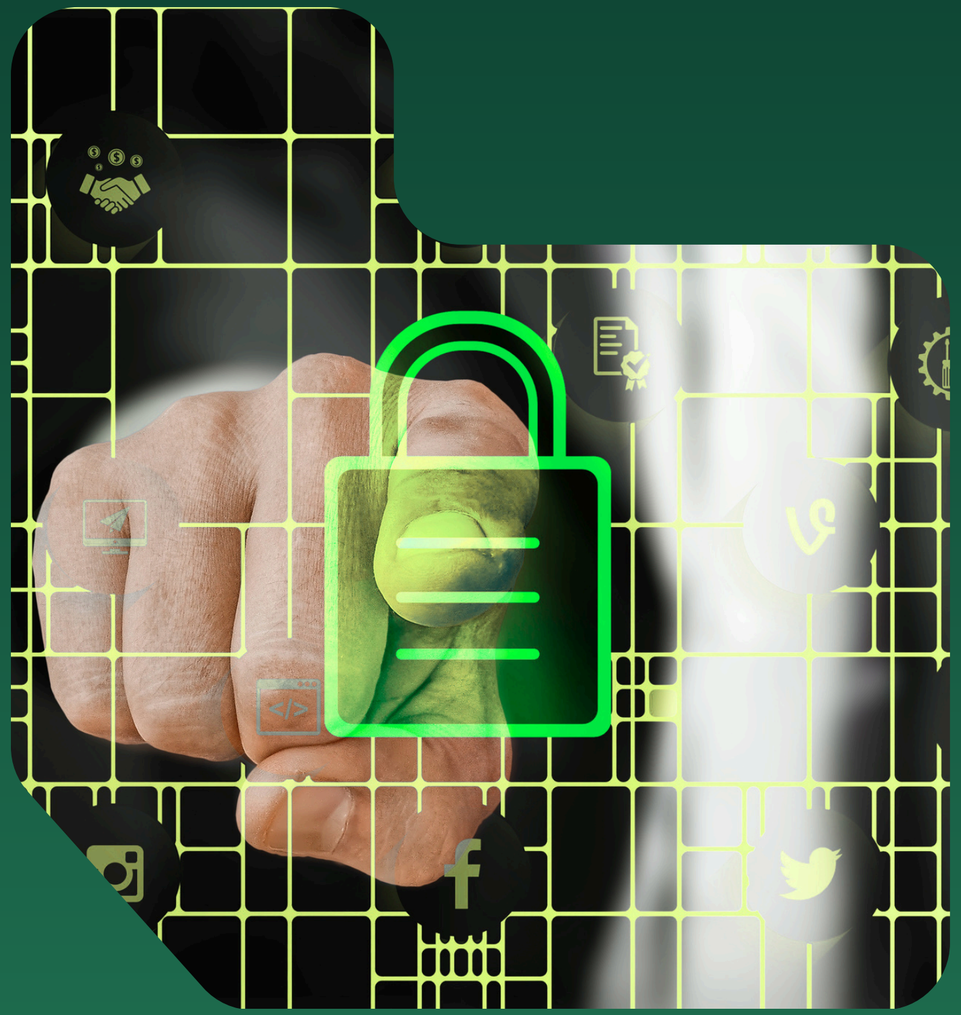
Sourashis Jana, roll–76

Saheb Adak. Roll– 47

FOREN SIGHTX



SUBJECT: Neural Network & Deep Learning (PCCCS587)



The Problem & Our Vision

Problem Statement

- Digital investigations involve large volumes of heterogeneous evidence
- Manual analysis is time-consuming and error-prone
- Different evidence types require different analysis techniques
- Investigators need explainable decision support
- Need for integrated multimodal evidence analysis

Project Overview

- AI-powered digital forensics intelligence assistant
- Analyzes chats, emails, images, and metadata
- Generates risk assessment and evidence timeline
- Produces explainable investigation reports
- Supports investigators, not legal decision-making

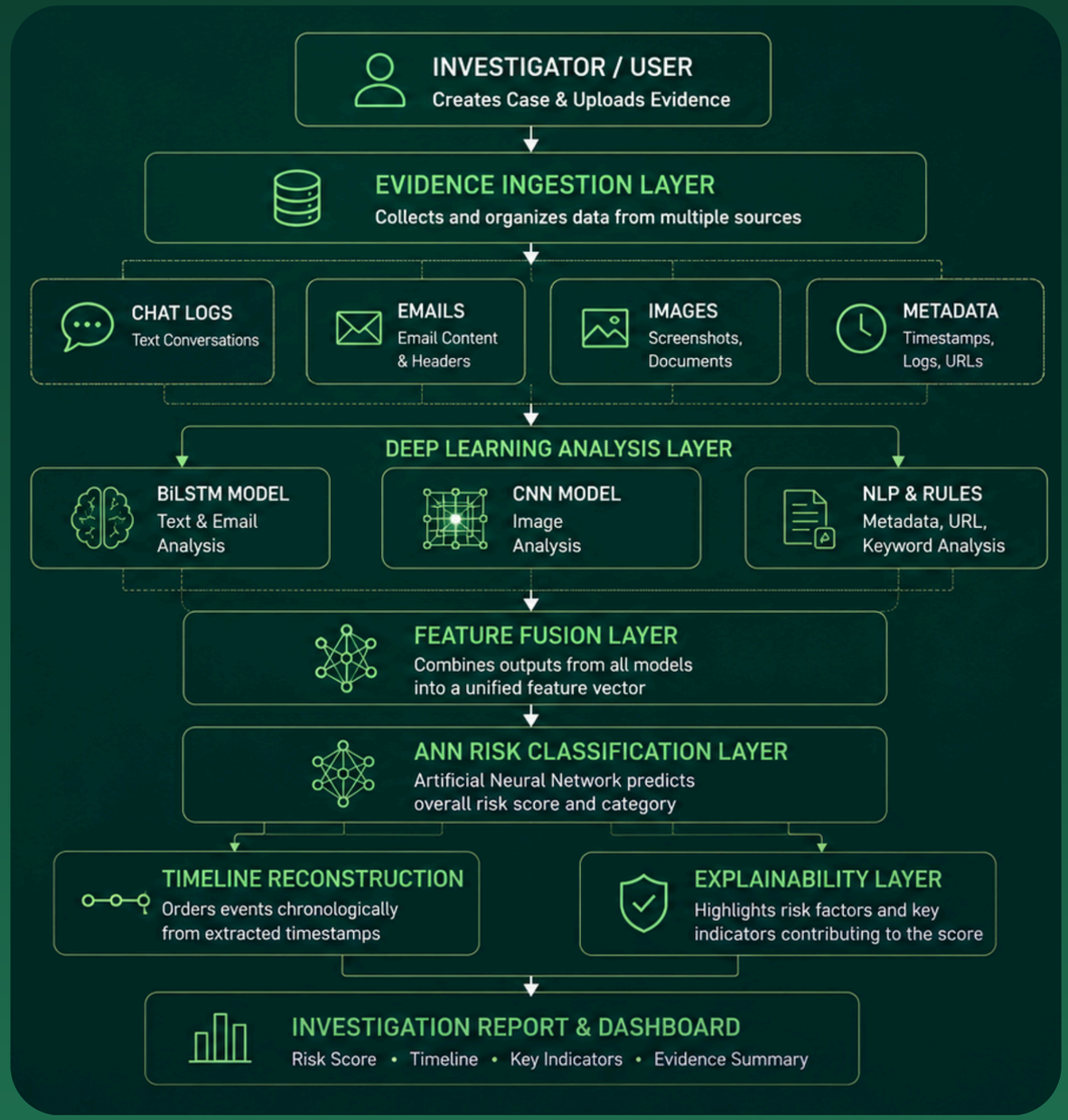
SYSTEM ARCHITECTURE

Architecture Pipeline

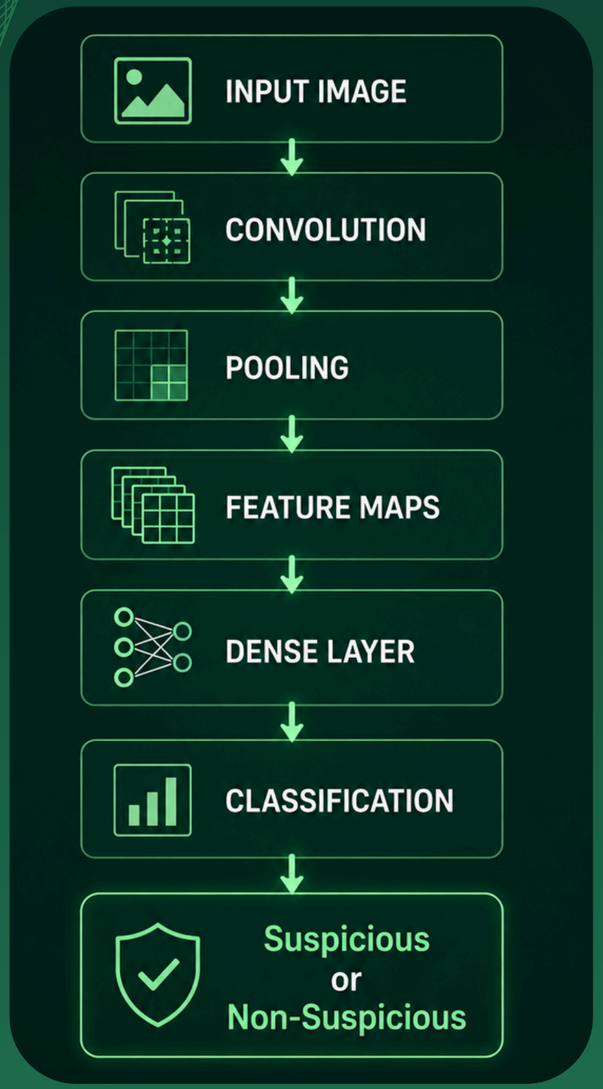
- Evidence ingestion from multiple sources
- Modality-specific preprocessing
- CNN for image analysis
- BiLSTM for text and email analysis
- ANN-based multimodal feature fusion
- Risk scoring, timeline generation, and reporting

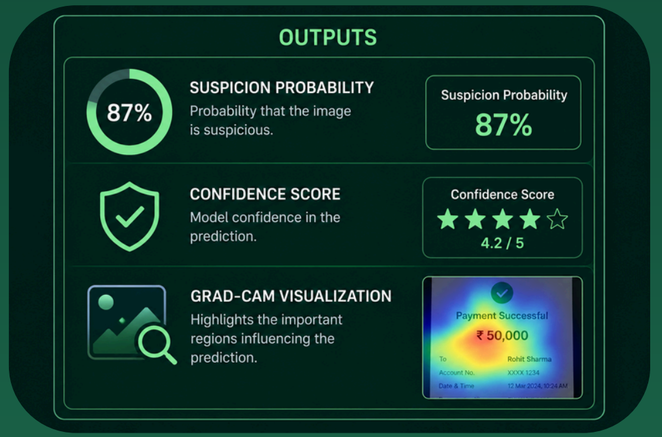
Core Components

- Evidence Ingestion Layer
- Deep Learning Analysis Layer
- Feature Fusion Layer
- Investigation Intelligence Layer
- Reporting Layer



CNN VISUAL EVIDENCE ANALYZER





Purpose

- Analyze screenshots and image evidence.
- Detect suspicious visual patterns.
- Identify manipulated images.
- Generate confidence scores.

CNN Pipeline

- Image Preprocessing
- Convolution
- Pooling
- Feature Extraction
- Classification

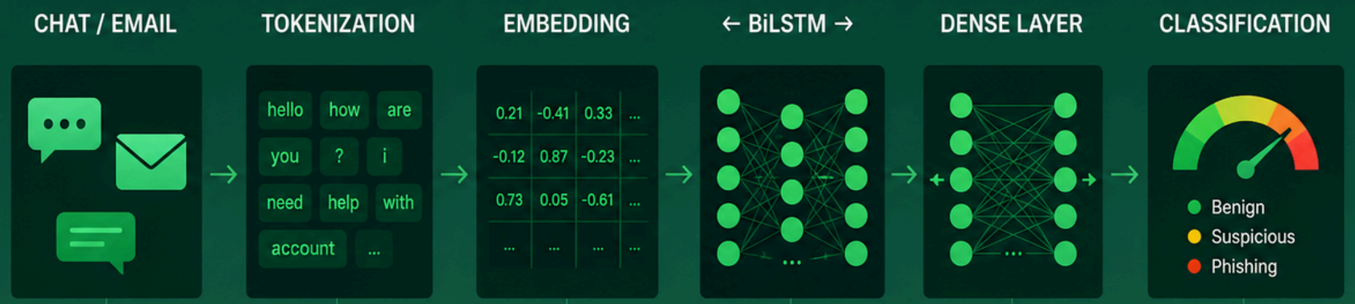
BiLSTM CONVERSATION INTELLIGENCE

Processing Steps

- Cleaning
- Tokenization
- Embedding
- BiLSTM
- Classification

Purpose

- Analyze chats and emails.
- Understand contextual relationships.
- Detect phishing and scam communication.

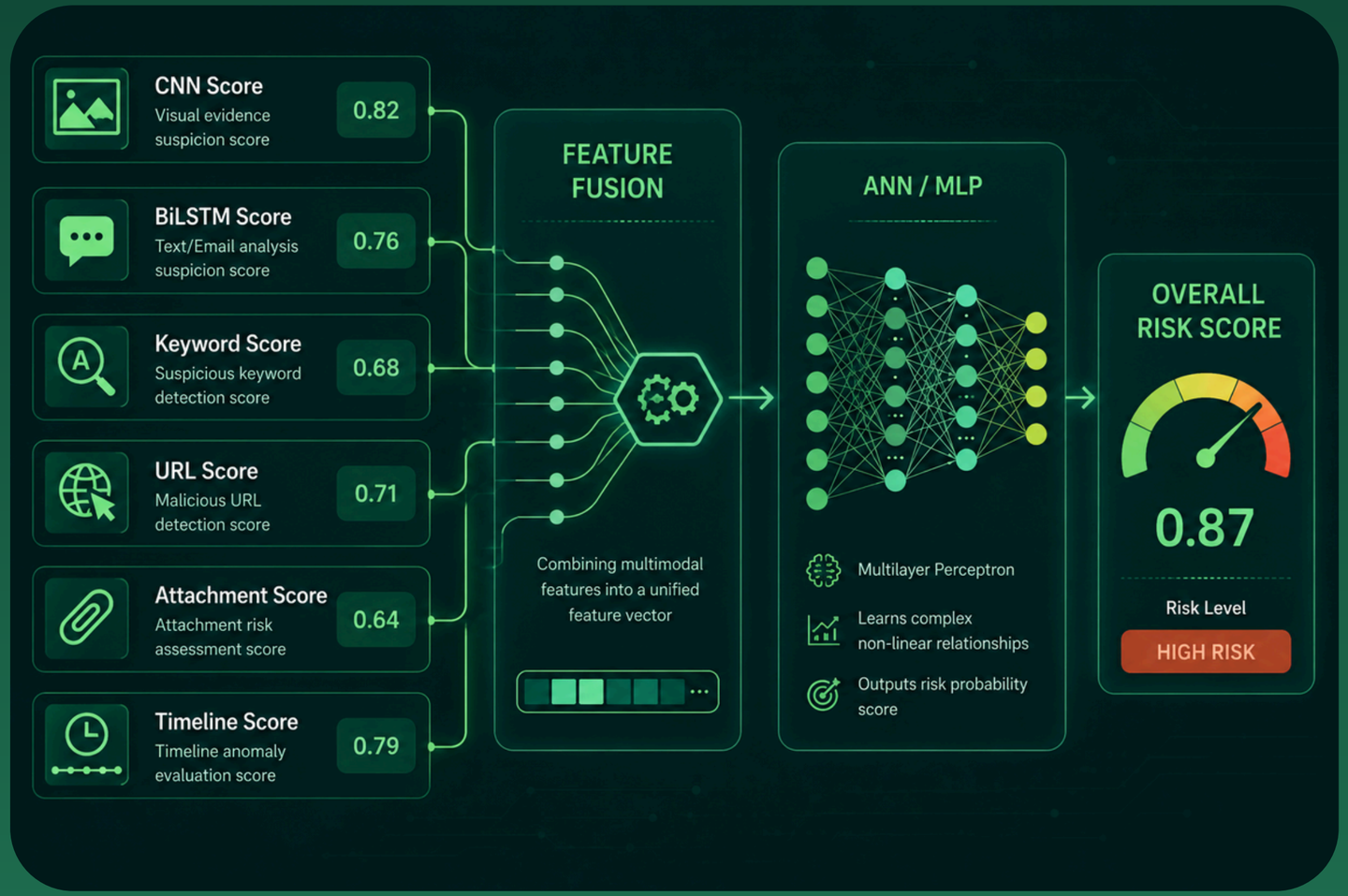


Why BiLSTM?

Reads text in both directions, gives better context understanding and suitable for sequence analysis.

Feature Fusion & ANN Risk Classification

Core Innovation of ForenSightX



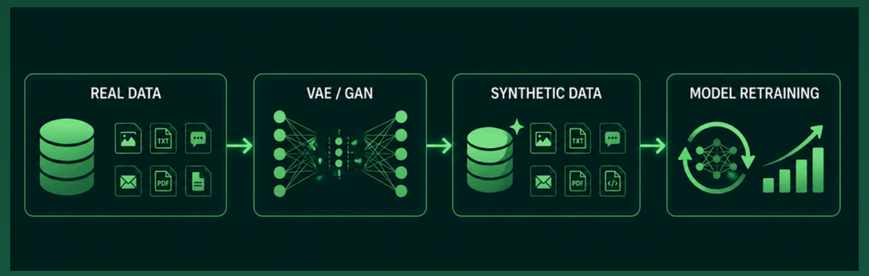
Input Features

- CNN Score
- BiLSTM Score
- URL Indicators
- Keyword Indicators
- Attachment Indicators
- Timeline Anomalies

ANN Output

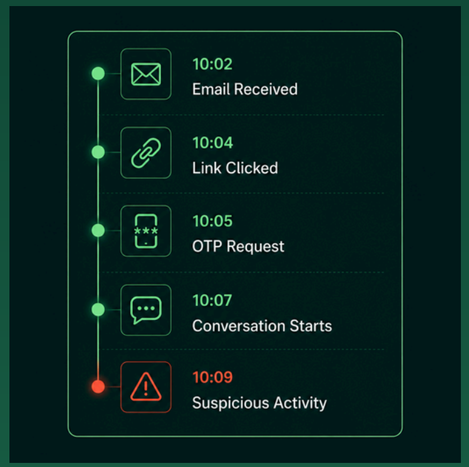
- Overall Risk Probability
- Investigation Support Score

VAE/GAN Augmentation, Timeline Reconstruction & Explainability



VAE/GAN Augmentation

- Generates synthetic training samples
- Addresses limited forensic datasets
- Enables controlled experimentation
- Compares baseline vs augmented training



Timeline Reconstruction:

- Extracts timestamped events
- Sorts events chronologically
- Reconstructs investigation sequence
- Supports incident analysis

Explainability Features

- Risk contributing indicators
- Grad-CAM visual explanations
- Important textual indicators
- Confidence score visualization

Technology Stack, Evaluation & Expected Outcomes

Technology Stack

Frontend:
- React
- Next.js
- Tailwind

Backend:
- FastAPI

DL:
- PyTorch

Database:
- SQLite/PostgreSQL

Evaluation Metrics

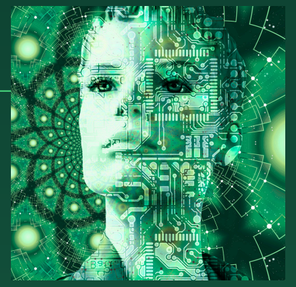
- Accuracy
- Precision
- Recall
- F1 Score
- ROC–AUC
- Confusion Matrix

Experiments

- A → Text Only
- B → Image Only
- C → Text + Image
- D → Text + Image + Metadata
- E → Multimodal + Augmentation

Outcomes

- Better multimodal classification
- Improved investigation support
- Explainable AI outputs
- Research contribution in evidence fusion



“THANK YOU”

